

THE WOLVERINE STACK

Research-Backed BPC-157 + TB-500 Protocol Guide

2026 Edition v3

Includes: Full Reconstitution Math & Mixing Tables

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Important Disclaimer

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This is educational information only. Not medical advice. Peptides discussed are for research purposes. Always consult your licensed physician before starting, stopping, or modifying any health regimen. Individual results vary. The information provided is not evaluated by the FDA.

What is the Wolverine Stack?

The "Wolverine Stack" combines two of the most researched healing peptides: BPC-157 (Body Protection Compound-157) and TB-500 (Thymosin Beta-4 fragment). Together, they are studied in research settings for accelerated recovery, tissue repair, and inflammation reduction.

This guide summarizes the current research as of 2026 — no bro-science, just the studies.

BPC-157 Deep Dive

What it is: A synthetic pentadecapeptide derived from a protective protein found in human gastric juice.

Key Mechanisms (2025-2026 Research)

- Promotes angiogenesis (new blood vessel formation) and collagen production
- Reduces systemic inflammation and modulates growth factors
- Protects and repairs the gut lining (cytoprotective effects)
- Supports tendon, ligament, and muscle healing

Research Highlights

Source	Finding
HSS Journal (2025)	Systematic review — BPC-157 improves outcomes in muscle, tendon, and ligament injuries in orthopaedic sports medicine.
Pharmaceuticals (2025)	Multifunctionality review — targets angiogenesis and nitric oxide pathways while maintaining protective functions.
Pilot Study (2025)	Intravenous BPC-157 demonstrated safety in humans.

Common Research Dosages: 250-500 mcg per day (split into 1-2 doses)

Best Research Applications: Gut health, tendon/ligament repair, inflammation reduction

TB-500 Deep Dive

What it is: A synthetic peptide fragment derived from Thymosin Beta-4 (Tβ4), an endogenous protein upregulated in injured tissues.

Key Mechanisms

- Promotes cell migration to injury sites and tissue restoration
- Reduces inflammation and supports muscle fiber repair
- Facilitates angiogenesis and wound healing

Research Highlights

Source	Finding
2026 study	TB-500 modulates cardiac remodeling by regulating ROCK1 expression.
2025 research	Low-temperature fabrication of Tβ4-loaded scaffolds accelerates wound healing via immune regulation.
2026 primer	Injectable peptide therapy shows promise in orthopaedic applications for tissue repair.

Common Research Dosages: 2.5-5 mg twice per week

Best Research Applications: Muscle recovery, injury healing, flexibility support

Stacking Protocols (Research-Based)

The BPC-157 + TB-500 combination is widely studied for synergistic healing effects.

Level	BPC-157	TB-500	Notes
Beginner	250 mcg 2x daily	2.5 mg 2x weekly	Good starting point
Intermediate	300 mcg 2x daily	5 mg 2x weekly	Most common research stack
Advanced	400-500 mcg 2x daily	5 mg 2x weekly	For experienced users only

Cycle Length: Most research protocols run 4-8 weeks, followed by a break.

Dosing Guide (Research Ranges Only)

Weight-Based Guidelines (Approximate)

- Under 150 lbs → Lower end of range
- 150-200 lbs → Standard research doses
- Over 200 lbs → May benefit from slightly higher end

Goal-Based Adjustments

- Recovery / Injury: Higher BPC-157 focus (300-500 mcg)
- Performance / Muscle: Balanced stack (Intermediate protocol)
- Longevity / General Wellness: Lower maintenance doses (Beginner protocol)

 DISCLAIMER

Always start low and assess tolerance. These are research ranges only — not medical recommendations.

Supplies & Equipment Checklist

As a nurse, I cannot stress this enough: the right supplies make the difference between a clean injection and a preventable infection. Here's exactly what you need before you start.

Required Supplies

Item	Specification	Why It Matters
Insulin Syringes	0.3 mL (30-unit) or 0.5 mL (50-unit) with 30G or 31G needle, 5/16" (8mm) length	Subcutaneous injections require short, ultra-fine needles. The 30G-31G gauge minimizes discomfort and tissue trauma. 0.3 mL syringes are ideal for peptide doses (typically 0.1-0.3 mL).
Alcohol Swabs	70% isopropyl alcohol, individually wrapped	The most critical tool for infection prevention. Use one fresh swab per injection site.
Bacteriostatic Water	Sterile water with 0.9% benzyl alcohol	Required for reconstituting lyophilized (freeze-dried) peptides. The benzyl alcohol inhibits bacterial growth, allowing multi-dose use.
Sharps Container	FDA-cleared, puncture-resistant, leak-proof	Never throw needles in household trash. A proper sharps container prevents needlestick injuries to you, your family, and sanitation workers.
Alcohol Prep Pads (extra)	For vial tops	Swab the rubber stopper of every vial before each draw — every single time.

Why Insulin Syringes Specifically?

- Gauge (30G-31G): The higher the gauge number, the thinner the needle. 30G is fine enough for comfortable subcutaneous injection while wide enough for peptide solutions to flow smoothly.
- Volume (0.3 mL or 0.5 mL): Most peptide doses are 0.1-0.3 mL. A 0.3 mL syringe gives you precise measurement at these tiny volumes.
- Length (5/16" / 8mm): Subcutaneous tissue sits just below the skin. A short needle ensures you inject into the fat layer, not muscle.

RN Tip: Buy boxes of 100 — they're inexpensive and you'll go through them faster than you think. Never reuse a syringe. Period.

Sterile Technique — What It Is & How to Do It

"Use sterile technique" is the most common advice given — and the least explained. Let me fix that.

What Is Sterile Technique?

Sterile technique is a set of practices used in healthcare to eliminate microorganisms from surfaces, equipment, and the injection field. The goal: prevent bacteria from entering your body through the injection site, the needle, or the vial.

In a hospital, sterile technique means gloves, drapes, and a controlled environment. For at-home peptide injections, the principle is the same but the practice is simpler: create and maintain a clean field.

Step-by-Step Sterile Technique for Peptide Injections

Step 1: Prepare Your Workspace

- Clear a clean, flat surface (counter, table, desk).
- Wipe the surface with an alcohol swab or disinfectant wipe.
- Lay out a clean paper towel as your "sterile field" — this is where your supplies go.
- Wash your hands thoroughly with soap and warm water for at least 20 seconds. Dry with a clean towel.

Step 2: Gather Supplies Without Contaminating Them

- Place your supplies on the clean paper towel: syringe (in wrapper until use), alcohol swabs, reconstituted peptide vial, sharps container nearby.
- Do NOT touch the needle or the rubber stopper of the vial with your bare fingers once you've washed your hands.

Step 3: Swab the Vial Top

- Tear open a fresh alcohol swab.
- Scrub the rubber stopper of the vial vigorously for 10–15 seconds.
- Let it air dry completely (takes about 15–20 seconds). Do NOT blow on it or fan it.

Step 4: Draw from the Vial

- Remove the syringe from its wrapper. Do not touch the needle tip or shaft.
- Pull the plunger back to draw air equal to your dose.
- Insert the needle through the center of the swabbed stopper at a 90° angle.
- Push the air into the vial (this pressurizes the vial and makes drawing easier).
- Invert the vial and draw your dose slowly. Tap the syringe gently to dislodge any air bubbles.

Step 5: Prepare the Injection Site

- Choose your site (see "Injection Sites" section below).
- Tear open a NEW alcohol swab (never reuse one).
- Scrub the injection site in a circular motion, starting from the center and spiraling outward.
- Scrub for 10–15 seconds covering an area about 2–3 inches across.
- Let it air dry completely. This is critical — injecting before the alcohol dries means it hasn't killed the bacteria yet AND it stings.

Step 6: Inject — Do Not Touch the Cleaned Site

- Once the site is cleaned and dried, do NOT touch it again.
- Pinch a fold of skin/fat at the injection site.

- Insert the needle at a 45–90° angle in one smooth motion.
- Inject slowly (about 5–10 seconds for a typical dose).
- Withdraw the needle at the same angle.
- Apply gentle pressure with a clean cotton ball or tissue — do NOT rub vigorously.

Step 7: Dispose Immediately

- Place the used syringe directly into the sharps container. Never recap the needle.
- Used alcohol swabs and wrappers go in regular trash.

RN Tip: The most common mistake I see is touching the cleaned injection site before injecting. Once that alcohol dries, the site is as clean as it's going to get. Don't undo your work.

Injection Sites & Technique

Subcutaneous Injection (Recommended for Peptides)

Subcutaneous means "under the skin" — injecting into the fatty tissue layer between skin and muscle. This is the standard route for peptide administration.

Best Injection Sites (Rotate Between Them)

1. Abdomen (stomach area): The easiest and most common site. Use the area 2 inches away from your belly button. Avoid the midline. This area has consistent fat thickness and good absorption.
2. Outer thigh: The front and outer area of the thigh, midway between hip and knee. Good if you prefer not to use your abdomen.
3. Back of upper arm: Harder to reach yourself but works if someone helps you.
4. Upper outer buttocks / hip: Good absorption, harder to self-administer.

RN Tip: Rotate sites daily. Injecting the same spot repeatedly causes scar tissue (lipohypertrophy), which reduces absorption and creates lumps under the skin. Use a simple pattern: left abdomen Monday, right abdomen Tuesday, left thigh Wednesday, etc.

Injection Angle

- Pinch an inch: Pinch a fold of skin and fat between your thumb and forefinger.
- 45° angle if you have less body fat (the needle needs to reach the fat layer without going into muscle).
- 90° angle if you have adequate body fat (pinched fold is 2+ inches).

Post-Injection Care

- Apply gentle pressure with a clean cotton ball for 10–15 seconds.
- Do NOT massage the site — this can accelerate absorption or irritate the tissue.
- A tiny drop of blood is normal. A small bruise occasionally happens and is harmless.

- If you notice redness, swelling, warmth, or pain that worsens over hours — seek medical attention.

Reconstitution & Mixing Math — Your Complete Guide

This is the section everyone asks for. How much bac water do I use? What concentration does that give me? How many units on the syringe equals my dose? Let's walk through it step by step.

RN Tip:

The math looks complicated the first time. It's not. Once you do it once, you'll never need the chart again. Until then — the charts are here.

Step 1: What You Need

Item	Specification	Why
Bacteriostatic Water	Sterile water with 0.9% benzyl alcohol	Required. The benzyl alcohol inhibits bacterial growth, allowing multi-dose use. Never use plain sterile water for multi-dose vials.
3 mL syringe (with needle)	22G-25G, 3 mL capacity	For transferring bac water INTO the peptide vial. The larger needle handles thick rubber stoppers.
Insulin syringe	0.3 mL (30-unit) or 0.5 mL (50-unit), 30G-31G	For injecting YOUR dose. Ultra-fine for subcutaneous comfort.
Alcohol swabs	70% isopropyl	One for the bac water vial stopper, one for the peptide vial stopper, one for your injection site. Three swabs minimum per session.

Step 2: How to Mix (Reconstitution Technique)

1. Wash your hands. At least 20 seconds with soap and warm water. Dry with a clean towel.
2. Swab both vial tops. The bac water vial AND the peptide vial. Scrub each stopper for 10-15 seconds with a fresh alcohol swab. Let air dry completely.
3. Draw air into the 3 mL syringe. Pull the plunger back to the volume of bac water you plan to add (e.g., 2 mL of air for 2 mL of water).
4. Inject air into the bac water vial. Insert the needle through the center of the stopper at 90°. Push the air in.
5. Draw the bac water. Invert the bac water vial with the needle still in it. Pull the plunger to your target volume.
6. Add water to the peptide vial — AIM AT THE GLASS. Aim the stream of water at the inside glass wall, NOT directly at the powder. Direct impact causes foaming.
7. Gently swirl. Do NOT shake. Shaking denatures the amino acid bonds. Swirl in small circles until fully dissolved (10-30 seconds).
8. Check clarity. The solution should be clear, not cloudy. Cloudiness = contamination or degradation. Discard if cloudy.

Step 3: Understanding Concentration

Once reconstituted, every vial has a concentration measured in mg/mL (milligrams per milliliter):

$$\text{Concentration (mg/mL)} = \text{Vial Size (mg)} \div \text{Bac Water Volume (mL)}$$

On an insulin syringe:

- A 0.3 mL (30-unit) syringe: each unit = 0.01 mL
- A 0.5 mL (50-unit) syringe: each unit = 0.01 mL
- A 1.0 mL (100-unit) syringe: each unit = 0.01 mL

$$\text{Your dose in mcg} = (\text{Units injected} \div 100) \times \text{Concentration (mg/mL)} \times 1000$$

Or the reverse — to find how many units you need:

$$\text{Units} = (\text{Desired Dose in mcg} \times \text{Bac Water mL}) \div (\text{Vial mg} \times 10)$$

Step 4: BPC-157 Mixing Tables

5 mg BPC-157 Vial (most common starter size)

Bac Water	Concentration	125 mcg	250 mcg	300 mcg	400 mcg	500 mcg
1.0 mL	5.0 mg/mL	2.5 units	5 units	6 units	8 units	10 units
2.0 mL	2.5 mg/mL	5 units	10 units	12 units	16 units	20 units
2.5 mL	2.0 mg/mL	6 units	12.5 units	15 units	20 units	25 units
3.0 mL	1.67 mg/mL	7.5 units	15 units	18 units	24 units	30 units

10 mg BPC-157 Vial (better value, lasts longer)

Bac Water	Concentration	125 mcg	250 mcg	300 mcg	400 mcg	500 mcg
1.0 mL	10.0 mg/mL	1.3 units	2.5 units	3 units	4 units	5 units
2.0 mL	5.0 mg/mL	2.5 units	5 units	6 units	8 units	10 units
3.0 mL	3.33 mg/mL	3.8 units	7.5 units	9 units	12 units	15 units
4.0 mL	2.5 mg/mL	5 units	10 units	12 units	16 units	20 units

RN Tip: For 10 mg BPC-157 vials, use 2 mL of bac water. This gives you 5 mg/mL — every 1 unit = 50 mcg. Easy math. The 250 mcg standard dose is exactly 5 units. Simple.

Step 5: TB-500 Mixing Tables

TB-500 doses are larger (typically 2.5–5.0 mg per injection), so reconstitution volumes matter less for precision. Still — here's the math.

5 mg TB-500 Vial

Bac Water	Concentration	1.0 mg	2.5 mg	5.0 mg
1.0 mL	5.0 mg/mL	20 units	50 units	100 units (full 1 mL)
2.0 mL	2.5 mg/mL	40 units	100 units	200 units (2 syringes)
2.5 mL	2.0 mg/mL	50 units	125 units	—

10 mg TB-500 Vial (most common)

Bac Water	Concentration	1.0 mg	2.5 mg	5.0 mg
1.0 mL	10.0 mg/mL	10 units	25 units	50 units
2.0 mL	5.0 mg/mL	20 units	50 units	100 units
3.0 mL	3.33 mg/mL	30 units	75 units	150 units

RN Tip:

For 10 mg TB-500 vials, use 2 mL of bac water. This gives you 5 mg/mL — your 2.5 mg dose = 50 units (0.5 mL), your 5.0 mg dose = 100 units (1.0 mL). Quick, clean math.

Step 6: The Formula (For Any Peptide, Any Dose)

Write this down. It's the only formula you'll ever need:

Units on syringe = (Desired dose in mcg ÷ Vial size in mg ÷ 1000) × Bac water in mL × 100

Example 1: 10 mg BPC-157 vial, 2 mL bac water, want 250 mcg

Units = (250 ÷ 10 ÷ 1000) × 2 × 100 = 0.025 × 2 × 100 = 5 units (correct)

Example 2: 5 mg TB-500 vial, 1 mL bac water, want 2.5 mg (2500 mcg)

Units = (2500 ÷ 5 ÷ 1000) × 1 × 100 = 0.5 × 1 × 100 = 50 units (correct)

Step 7: Storage After Mixing

State	Temperature	Shelf Life	Notes
Dry powder (unmixed)	Cool, dry place or refrigerated	12-36 months (per manufacturer)	Keep away from light and moisture
Reconstituted (mixed)	2-8°C (36-46°F) refrigerated	~30 days (use within 1 month)	Do NOT freeze. Keep vial upright.
In syringe	Refrigerated	Use within 24 hours	Not recommended for routine use
Traveling	Insulin case with ice pack	Duration of travel	Never leave peptides in a hot car

⚠️ DISCLAIMER

Reconstituted peptides degrade over time. Mark your vial with the date you mixed it. Discard any unused solution after 30 days — the benzyl alcohol preservative slows bacterial growth but doesn't stop it indefinitely.

Safety, Contraindications & Monitoring

Who Should Exercise Caution

- History of cancer (angiogenesis concerns with BPC-157 — promotes blood vessel growth)
- Pregnancy or breastfeeding (no safety data available)
- Known allergies to peptides or any component
- Bleeding disorders or taking anticoagulants (BPC-157 affects blood vessel formation)

Recommended Monitoring

- Baseline blood work: CBC (complete blood count), CMP (comprehensive metabolic panel), CRP (inflammatory marker)
- Follow-up labs at 4-6 weeks into your protocol
- Track changes: energy, sleep, pain levels, recovery time — write them down

Red Flags — Stop Immediately and Seek Medical Care

- Unusual swelling, redness, or warmth at injection site
- Severe pain that worsens over time
- Signs of allergic reaction: rash, itching, difficulty breathing, swelling of face/throat
- Fever or flu-like symptoms after injection

Sources & References (2025-2026)

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Next Steps

Your next options:

1. Re-run the calculator anytime at nurserobhealth.com
2. Book a 1-on-1 Peptide Protocol Review with Nurse Rob, RN — \$197 for 30 minutes. Full stack safety review + personalized plan.
3. Join the 90-Day Peptide Mastery Group — \$997/quarter. Monthly group calls, private Discord, community accountability.

Book here: nurserobhealth.com

Stay safe and keep learning,

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